CTRL SYNC

Adaptive Gaming Controller Attachment | GNG 2101 E | Group 3.1

PROBLEM STATEMENT

The Problem

Standard controllers require two hands, locking out one-handed gamers from most titles.

Our client wants to use a Nintendo switch independently. Existing solutions can't be operated with one hand, and a custom controller is far too expensive.

The Goal

Design a simple, adjustable add-on attachment for an existing Nintendo Switch controller that is easy to use and improves accessibility.

DESIGN FOR X

1 **Accessibility**
Enable the client to operate both joysticks, triggers, and bumpers using only his right hand

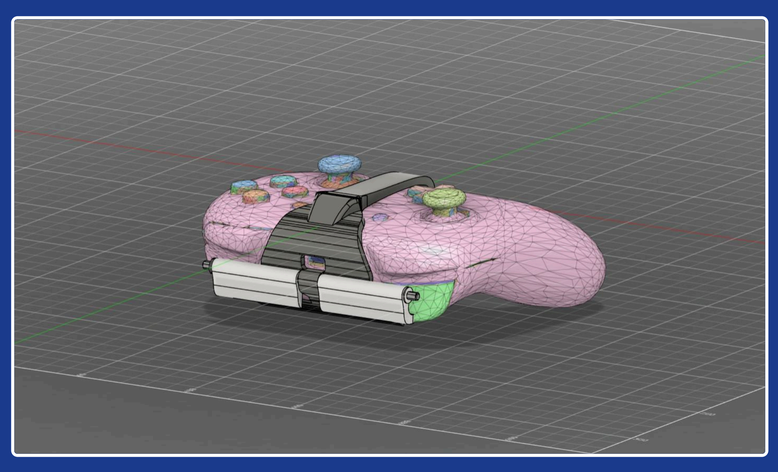
2 **Assembly**
Attachment can be installed, removed, and adjusted without tools or assistance

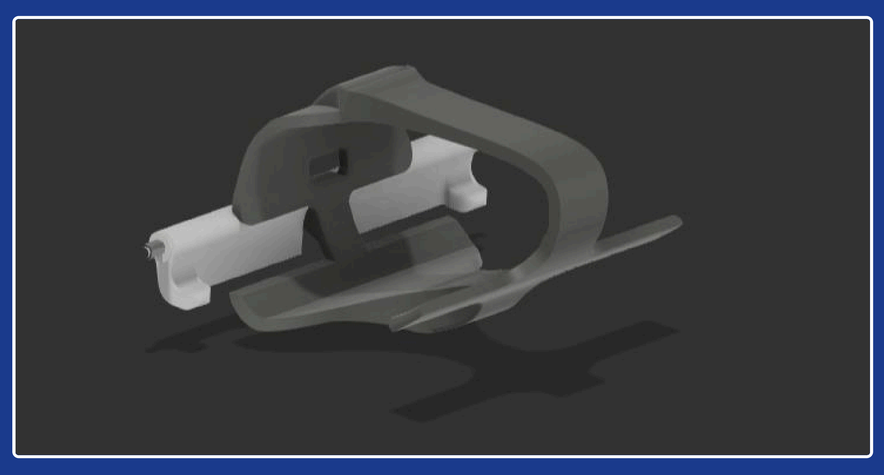
3 **Reliability**
Device functions consistently and without failure during sessions of 3+ hours

4 **Sustainability**
Minimize material waste by using recyclable or reusable components where possible

5 **Cost**
Keep the total cost of materials and manufacturing under $50

DESIGNS





USER NEEDS

High

- Compatible with existing controller
- One handed access to all controls
- Independent use
- Easy to learn and use
- Easy to install and remove
- Comfortable for extended use

Medium

- Does not block screen or restrict movement
- Lightweight with minimal weight added on
- Minimal noise during use

Low

- Portable and easy to carry
- Visually aesthetic
- Compact when not in use

CLIENT MODEL

About the Client

Our client is a male individual with full use of his right hand only. He owns a Nintendo Switch and wants to play independently using his existing controller, without assistance or expensive alternatives. He values a solution that is simple, reliable, and works with the way he holds the controller.

ECONOMICS

1 **Gross Profit**

Sales Revenue: $4,380,000

Cost of Goods Sold: $25,900,000

Gross Profit: ($21, 520, 000)

2 **Expenses & Taxes**

Operating Expenses: $49, 200, 00

Operating Income: (49, 200, 000)

Financial Cost: $81, 00

3 **Net Income**

Earnings Before Taxes (EBT): ($49, 281, 000)

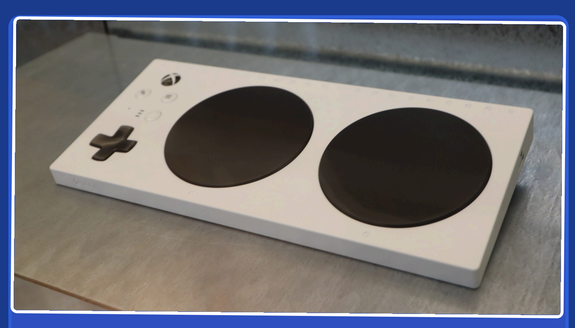
Taxes: Income Tax Expense (less: running deficit)

Total: ($49, 281, 000)

SPECIFICATIONS

#	Client's Need	Metric	Unit
1	Access all triggers	Max trigger actuation force	N
2	Reach all controls	Effective trigger reach extension	mm
3	Comfortable long-term use	Material comfort & surface texture rating	1–5 scale
4	Easy installation	Installation time	sec
5	Independent setup	Tools required	Y/N
6	Affordability	Manufacturing cost	$

BENCHMARKING



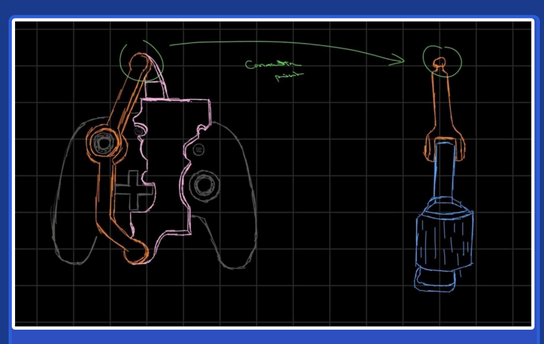
XBOX Adaptive

Joystick Attachment

BILL OF MATERIALS

Item Name	Description	Cost	Type
Magnets	For connecting the clipping attatchment, improving durability and accesibility.	$3.50	Material
ABS Filament	Used for the entire body of the attatchment, also used for the triggers to make it more accesible.	$1.30	Material
Fusion	Fusion is a prototyping software that allows users to utilize CAD to design projects.	Free	Software
Nuts	Nuts used in original prototypes as a replacement for the function of the magnet.	$0.10	Material
Total	-	**$4.90**	-

TESTING & PROGRESSION



Initial Sketch

First CAD

#	Metric	Unit	Ideal
1	Max trigger force	N	6.0 ± 0.5 N
2	Trigger reach extension	mm	30 – 45mm
3	Comfort & texture rating	1–5 scale	≥ 3.75/5
4	Installation time	sec	≤ 90 sec
5	Tools required	Y/N	No
6	Manufacturing cost	$	$45 – $50